Complete Assignment Answers (Questions 1-50)

Q1. A variable is a named memory location used to store data.

Q2. Variables store values that can change during program execution.

Q3. Variable name is identifier; value is stored data.

Q4. b) total_marks

Q5. C++ is case-sensitive.

Q6. No, variable names cannot contain spaces.

Q7. Yes.

Q8. Compilation error in same scope.

Q9. studentName, studentAge, studentMarks

Q10. studentMarks because it is meaningful.

Q11. A data type defines the kind of data stored.

Q12. int

Q13. float or double

Q14. double has higher precision.

Q15. char

Q16. bool

Q17. string

Q18. 18

Q19. 25=int, 3.14=float/double, 'A'=char, true=bool

Q20. Decimal part is truncated.

Q21. To take input from user.

Q22. To display output.

Q23. >>

Q24. <<

Q25. Inputs value into age.

Q26. Displays value of age.

Q27. Moves cursor to next line.

Q28. cin >> marks;

Q29. cout << "Welcome to C++";

Q30. Programming

Q31. A symbol that performs operations.

Q32. +

Q33. 1

Q34. 0

Q35. ==

Q36. = assigns, == compares.

Q37. true

Q38. Logical AND

Q39. Logical OR

Q40. Logical NOT

Program Based Questions (41-50)

41. Hello World

Problem Statement: Print Hello World

Algorithm: Read input, process, display result.

```
#include <iostream>
using namespace std;
int main(){
 cout<<"Hello World";
 return 0;
}
```

42. Sum of Two Numbers

Problem Statement: Input two numbers and display sum

Algorithm: Read input, process, display result.

```
#include <iostream>
using namespace std;
int main(){
 int a,b;
 cin>>a>>b;
 cout<<"Sum = "<<a+b;
 return 0;
}
```

43. Area of Rectangle

Problem Statement: Calculate area

Algorithm: Read input, process, display result.

```
#include <iostream>
using namespace std;
int main(){
 int l,b;
 cin>>l>>b;
 cout<<"Area = "<<l*b;
 return 0;
}
```

44. Formula

Problem Statement: Simple Interest = (P*R*T)/100

Algorithm: Read input, process, display result.

45. Simple Interest Program

Problem Statement: Calculate SI

Algorithm: Read input, process, display result.

```
#include <iostream>
using namespace std;
int main(){
 float p,r,t;
 cin>>p>>r>>t;
 cout<<(p*r*t)/100;
 return 0;
}
```

46. Temporary Variable

Problem Statement: Used while swapping

Algorithm: Read input, process, display result.

```
temp=a; a=b; b=temp;
```

47. Swap Numbers

Problem Statement: Swap a and b

Algorithm: Read input, process, display result.

```
int a=10,b=20,temp;
temp=a;
a=b;
b=temp;
```

48. Formula

Problem Statement: F=(C*9/5)+32

Algorithm: Read input, process, display result.

49. Fahrenheit Value

Problem Statement: 0°C = 32°F

Algorithm: Read input, process, display result.

50. Integer Division

Problem Statement: 9/5 gives 1; use 9.0/5

Algorithm: Read input, process, display result.